

RESEARCH ARTICLE

Super-Foldable Glass-Like Plastic

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ABSTRACT

Fatigue-free foldability is required for next-generation flexible electronics. For the essential protective films, ultrathin glass is susceptible to brittle fractures, whereas plastic films have inadequate hardness and tend to crease under dynamic loading conditions. We report a glass-like plastic featuring a nano-hybrid interpenetrating network comprising a plastic nanofibrous scaffold interpenetrated by an organic-inorganic silsesquioxane@nanosilica composite, which effectively inhibits fatigue relaxation. This hybrid material possesses glass-like transparency and hardness yet exhibits plastic-like elongation and impact resistance, and rubber-like resilience. When manufactured as thin films (5–30 μm), they can withstand extreme folding cycles (500,000 cycles at a radius of curvature of 0.5 mm) without macroscopic creasing/cracking or microscopic structure/morphology changes. The superior dynamic foldability of these glass-like plastics makes them promising for future device applications.

1 | Introduction

Since the invention of telephones, electronic communication devices have undergone significant technological evolution, having transitioned from stationary to mobile forms, and are now advancing toward flexible and even foldable devices (Figure S1) [1–9]. Regardless of the electronic functions, the active components need to be integrated/packaged with a protective film. The key challenge in designing mechanically deformable electronics lies in developing protective films that seamlessly integrate glass-like optical transparency and surface hardness with plastic-like pliability. Importantly, the material employed must exhibit crease- and crack-resistant foldability under harsh dynamic loading conditions [10–12]

Conventional plastic films (e.g., colorless polyimide, CPI) exhibit flexibility and toughness but suffer from inadequate hardness (<0.3 GPa) [13–15], necessitating protective hard coatings (Figure S2A). Utilizing branched, ladder-like, cage-like polysiloxanes

with a rigid inorganic core and organic periphery has prepared a series of hard yet flexible coatings [13, 16–21]; however, these laminated structures tend to develop irreversible creasing under cyclic loading conditions due to the inherent fatigue relaxation of the plastic substrates [22, 23]. This issue is particularly pronounced for foldable displays, which are required to sustain many cycles of mechanical loading at a radius of curvature (r_c) < 1.0 mm. Ultrathin glass (UTG), with a thickness of 30 to 50 μm , has emerged as an attractive alternative owing to its high surface hardness (>7 GPa) and modulus (>60 GPa), which impart excellent scratch resistance and mitigate creep [24, 25]. However, UTG's intrinsic brittleness renders it susceptible to catastrophic brittle fracture under impact [25, 26], necessitating the use of additional plastic films, scratch-resistant coatings, or soft protective layers to ensure adequate durability (Figure S2B) [27]. While bare UTG can achieve a r_c of 0.5–1 mm [27, 28], the system-level foldability is ultimately constrained by these protective layers, which are prone to creasing and compromise the device's aesthetic quality. Recently, hybrid films have been developed by impregnating ultraviolet (UV) curable hybrid polymer (e.g., oligosiloxane) into

electrospun nanofibers [29] or phase-inversion porous films [30], providing valuable insights into the design of flexible protective films. Nevertheless, further advances toward achieving super-foldability under more demanding conditions ($>200,000$ cycles at $r_c < 1$ mm) remain an important direction for next-generation display applications. The critical challenge lies in reconciling the inherent trade-off between high hardness, impact resistance, and extreme foldability and, more importantly, mitigating crease and crack formation while maintaining superior optical transparency and facile manufacturability.

In this study, we develop a class of glass-like plastics (GLPs) featuring a nano-hybrid interpenetrating network, composed of a porous ultrahigh molecular weight polyethylene (PE) nanofibrous scaffold as the soft phase and a highly crosslinked 3-glycidyloxypropyl functionalized oligomeric silsesquioxane@nanosilica (GOS@nanosilica) composite as the hard phase (Figure 1A–C). The porous PE nanofibrous scaffold consists of highly oriented nanofibrils, offering excellent mechanical properties. Meanwhile, the clear and viscous GOS@nanosilica composite, with tunable nanosilica content and facile processability, matches the refractive index of PE fibrils. Upon UV curing, it exhibits transparency and provides high hardness and modulus, effectively inhibiting plastic relaxation and fatigue failure. Notably, the synergistic effects of organic-hybrid-inorganic components and nano-hybrid interpenetrating network result in materials with high transparency (92%, 550 nm), glass-like hardness (1.1 GPa, 8–9H), plastic-like elongation ($\epsilon_b > 6\%$), rubber-like elastic recovery (78.3%), and superior impact resistance. Furthermore, due to the facile processability, tunable porosity (20%–80%), and controllable thickness (10 μm –20 nm) of the PE scaffold [31, 32], such materials can be fabricated as thin films with a thickness of 5–30 μm and can withstand extreme folding cycles ($>500,000$ cycles at a r_c of 0.5 mm) without macroscopic creasing/cracking or microscopic structure/morphology changes (Figure S2C). We believe our materials transcend conventional boundaries between soft and hard substances, and their super-foldability makes them a promising candidate for next-generation protective films in flexible electronics.

2 | Results

2.1 | Fabrication of GLP

The fabrication of GLP involves the infiltration of a GOS@nanosilica dispersion into a pre-formed porous PE nanofibrous scaffold, followed by pressing and UV curing to construct a nano-hybrid interpenetrating network (Figure 1A–C). We first prepared the porous PE nanofibrous scaffolds through a well-established three-step process: melt-extrusion of ultrahigh molecular weight PE/petrolatum blends, biaxial stretching, and solvent extraction [31, 32]. This method yields freestanding scaffolds with controlled thicknesses of 0.8, 2, and 3 μm , designated PE-0.8, PE-2, and PE-3, respectively (Figure S3). Scanning electron microscopy (SEM) and atomic force microscopy (AFM) images (Figure S4) show that the scaffolds have porous shish-kebab crystalline nanofibrous structures with an average fibril diameter of 30–50 nm, much smaller than the wavelength range of visible light (380 to

780 nm). The GOS@nanosilica dispersion comprises three key components: molecularly engineered GOS, nanosilica, and a photoinitiator. GOS, the main component, includes cage-like, incomplete cage-like, and ladder-like 3-glycidyloxypropyl silsesquioxanes, with molecular weights ranging from 1300 to 2100 g mol^{-1} and a polydispersity index of 1.84 (Figure 1B and Figures S5 and S6A–E). Such structural diversity is typical for sol-gel-derived silsesquioxane systems and arises from the inherent complexity of the condensation process. The siloxane framework of GOS shows a 97.9% degree of condensation, as determined by liquid-state ^{29}Si NMR spectroscopy (Figure S6C). Nanosilica particles, averaging 18.5 nm in diameter (Figure S6F,G), are dispersed in situ within the GOS matrix, forming a stable, clear dispersion (Figure S6H,I) with good fluidity at elevated temperature (Figure S6J–L). Due to the closely matched surface energies of PE and GOS@nanosilica (Figure S7), the dispersion readily infiltrates the porous PE scaffold. After pressing and UV curing for 10 min, GLPs (~ 5 μm , Figure S3) with varying PE nanofibrous scaffold thicknesses (GLP-0.8, GLP-2, and GLP-3) are obtained via an effective cationic photopolymerization process. Fourier transform infrared (FT-IR) spectra demonstrate that the conversion of epoxy groups reaches 89% and remains unchanged after 24 h ambient storage (Figure S8), indicating efficient curing. The incorporation of nanosilica slightly reduces the refractive index of GOS from 1.53 to 1.51 at 550 nm, thereby moderating the overall refractive index of the hybrid matrix and reducing the refractive index contrast with the PE phase (Figure S9A). In addition, considering that both nanosilica particles and PE nanofibrils possess characteristic dimensions well below the wavelength of visible light, light scattering in the GLP system falls within the Rayleigh regime, where scattering intensity scales strongly with particle size, resulting in intrinsically weak scattering. The uniform dispersion of these nanoscale components further minimizes large-scale refractive index fluctuations, thereby suppressing the transition to stronger scattering regimes (e.g., Mie scattering). These combined effects result in GLPs with high optical transparency (92%, 550 nm) and low haze ($< 1\%$), comparable to that of inorganic UTG over the visible wavelength range (Figure 1D–F, Figure S10, and Table S1). Meanwhile, the in-plane refractive indices (at 0° , 45° , and 90°) of GLP-3 exhibit identical values, which are also very close to the out-of-plane refractive index (Figure S9B), indicating that the nano-hybrid interpenetrating network exhibits no birefringence and would not degrade the optical properties of GLP in flexible displays.

The surface morphology of GLP, characterized by SEM, reveals consistently smooth surfaces across different porous PE nanofibrous scaffold thicknesses (Figure S11A–C). The integration of the porous PE nanofibrous scaffold and the cured GOS@nanosilica (CGN) was confirmed using FT-IR spectroscopy, solid-state ^{29}Si NMR spectroscopy, and wide-angle X-ray diffraction (WAXD). The FT-IR spectrum of GLP-3 shows characteristic Si–O–Si vibrations from CGN, $-\text{CH}_2$ vibrations from PE fibrils, and $-\text{OH}$ vibration from the 3-glycidyloxypropyl group after photocuring (Figure S11D). Solid-state ^{29}Si NMR spectrum confirms the coexistence of T-unit GOS and Q-unit nanosilica (Figure S11E). Meanwhile, WAXD results indicate the successful integration of the amorphous CGN phase with the crystalline PE nanofibrous scaffold (Figure S11F).

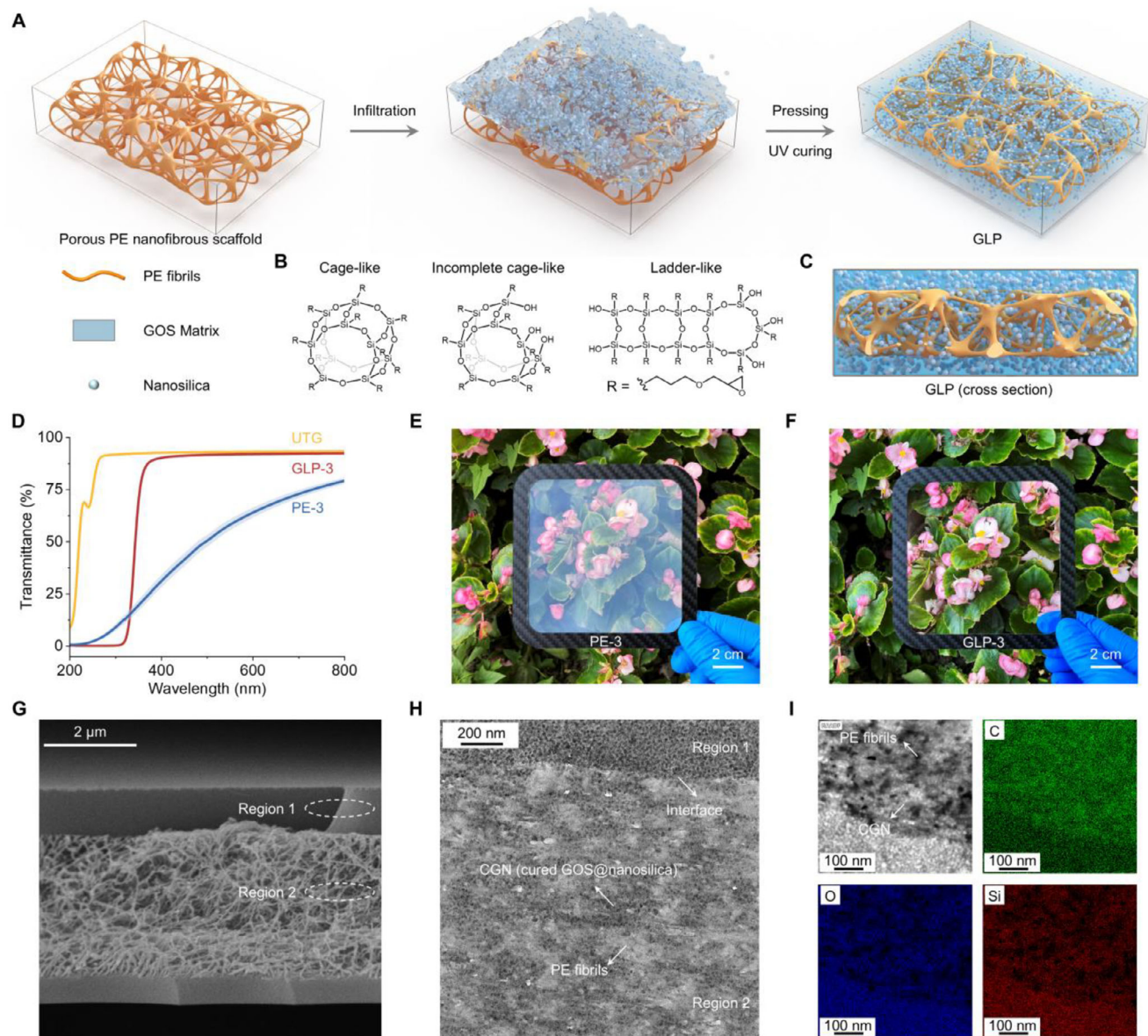


FIGURE 1 | Design and characteristics of GLP. (A) Scheme of the GLP preparation process. (B) Representative molecular structures of the GOS, including cage-like, incomplete cage-like, and ladder-like structures. (C) Schematic cross-sectional illustration of GLP-3. (D) Transmittance spectra of UTG, GLP-3, and PE-3 across a wavelength range of 200–800 nm. (E, F) Photographs of PE-3 (E) and GLP-3 (F) (100 cm²). (G) SEM image of the GLP-3 cross-section showing porous PE nanofibrous scaffold symmetrically interpenetrated within CGN. (H) TEM images of region 2 in the GLP-3 cross-section. (I) HAADF-STEM image of GLP-3 and the corresponding EDS 2D elemental mappings of C, O, and Si.

The nano-hybrid interpenetrating network structure of GLP was further confirmed by SEM and transmission electron microscopy (TEM). Cross-sectional SEM image shows symmetrical interpenetration of the porous PE nanofibrous scaffold within the CGN matrix, showing sandwich-like structure (Figure 1G and Figure S12). It should be noted that the cross-sectional morphology shows features of fibril pull-out from the highly oriented PE nanofibers during sample preparation, which can give rise to a filamentous appearance rather than indicating unfilled pores. Specifically, the sandwich-like structure comprises two distinct regions: the surface CGN phase (region 1) and the internal interpenetrating phase of CGN and porous PE nanofibrous scaffold (region 2), with CGN present in both regions (Figure S13A). Bright-field TEM, high-angle annular dark-field scan-

ning transmission electron microscopy (HAADF-STEM), and corresponding energy-dispersive X-ray spectroscopy (EDS) elemental mapping of C, O, and Si provide more direct evidence of uniform nanoscale infiltration, confirming effective nanoscale interpenetration of the amorphous hybrid matrix with PE fibrils (Figure 1H,I). The elemental distribution indicates that the inorganic phase is homogeneously present throughout the scaffold. By leveraging contrast differences between PE and CGN phases in HAADF-STEM images, the volume fraction of PE fibrils in region 2 is calculated as 76.8% (Figure S13B). These results underscore the crucial role of hybrid GOS in interpenetrating the organic PE nanofibrous scaffold with rigid inorganic nanosilica and, more importantly, verify the formation of a nano-hybrid interpenetrating network in GLP.

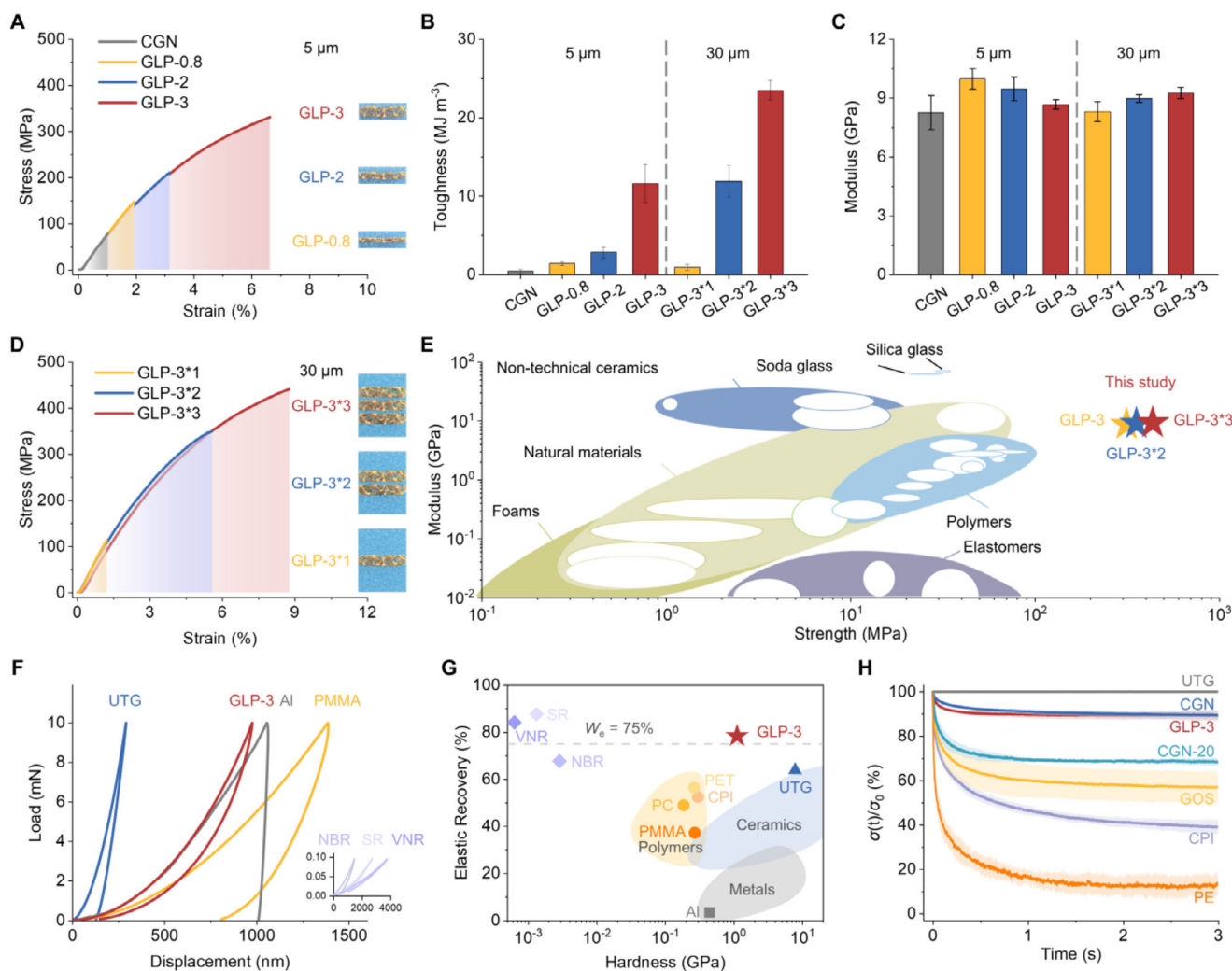


FIGURE 2 | Mechanical properties of GLP. A Representative stress–strain curves of CGN and GLPs (5 μm) with varying porous PE nanofibrous scaffold thicknesses. Insets are schematic cross-sectional illustration of GLP-0.8, GLP-2, and GLP-3. B–C Toughness (B) and modulus (C) from tensile tests of CGN and GLPs. D Representative stress–strain curves of GLP-3*1, GLP-3*2, and GLP-3*3 (30 μm). Insets are schematic cross-sectional illustration of GLP-3*1, GLP-3*2, and GLP-3*3. E Ashby plot of modulus and strength of GLPs, natural materials, and typical synthetic materials, including non-technical ceramics, polymers, foams, and elastomers. Reproduced with permission [36]. Copyright 2011, Elsevier. F Representative load–displacement curves for GLP-3 and typical synthetic materials, including ceramics, metals, polymers, and rubbers. Al, aluminum; PMMA, polymethyl methacrylate; SR, silicone rubber; NBR, nitrile butadiene rubber; VNR, vulcanized natural rubber. G Plot of elastic recovery vs. H for Al, PMMA, PC, PET, CPI, UTG, SR, NBR, VNR, GLP-3, and other engineering materials, including ceramics, metals, polymers, and rubbers. Reproduced with permission [37]. Copyright 2023, Springer Nature. H Normalized stress relaxation curves of PE, CPI, GOS, CGN-20 (with 20 wt.% of nanosilica), CGN (with 30 wt.% of nanosilica), GLP-3, and UTG, the input force (~ 10 nN) applied in ~ 0.1 s at room temperature.

2.2 | Mechanical Properties of GLP

Owing to their interpenetrating nano-hybrid interpenetrating network structures and the integration of organic, hybrid, and inorganic chemical components, GLP exhibit an unusual combination of plastic-like deformability, glass-like hardness, and rubber-like resilience. These mechanical properties were first evaluated through tensile tests. CGN behaves like brittle materials (e.g., glass and ceramics), with ϵ_b ranging from 0.5 to 1.5% (Figure 2A). In contrast, porous PE nanofibrous scaffolds show ductile behavior with ϵ_b exceeding 10% for PE-0.8 and reaching up to 35% for PE-3. Moreover, these porous scaffolds exhibit high strength (70–200 MPa), modulus (1–4.3 GPa), and toughness

(15–25 MJ m^{-3}), ensuring a well-balanced combination of strength and toughness (Figure S14). In fact, when porosity is accounted for, the intrinsic mechanical properties of the internal nanofibers within these PE scaffolds are expected to be very high. The incorporation of a porous PE nanofibrous scaffold significantly enhances GLP's ductility compared to brittle materials such as glass and ceramics (Figure 2A and Figure S15A). GLP's strength, ϵ_b , and toughness increase with the thickness of the incorporated porous PE nanofibrous scaffold (Figure 2B and Figure S16A,B). Specifically, the GLP-3 (5 μm) incorporating the PE-3 scaffold achieves a strength of 315.3 MPa, which is 4.2 times that of CGN. Additionally, its ϵ_b and toughness improve approximately 5.3 fold (6.1%) and 27.3 fold

(11.6 MJ m⁻³), respectively, while retaining CGN's high modulus (8.2 GPa) (Figure 2C). We attribute the excellent mechanical properties of the GLP to the synergistic effect between the crosslinked CGN networks and the highly stretched ultrahigh molecular weight PE nanofibrous scaffold, which is recognized as one of the strongest plastics. [32, 33] Moreover, a higher volume fraction of the PE scaffold in the 5 μ m GLP correlates with enhanced mechanical properties. Building upon this strategy, GLPs with a thickness of 30 μ m were fabricated by pressing one, two, or three layers of the PE-3 nanofibrous scaffold, designated GLP-3*1, GLP-3*2, and GLP-3*3, respectively. The synergistic effect between the crosslinked CGN networks and the PE nanofibrous scaffolds also enhances the mechanical properties of GLP-3*1, GLP-3*2, and GLP-3*3 (Figure 2B–D, Figures S15B, S16A, and S16B). Notably, increasing the number of PE nanofibrous scaffold layers amplifies reinforcement. GLP-3*3 exhibits a high modulus of 8.6 GPa, along with the highest strength (436.8 MPa), elongation at break (8.6%), and toughness (23.5 MJ m⁻³), while maintaining high optical transparency (Table S1 and Figure S16C). Post-strain SEM images of GLP-3*3 (Figure S17) reveal that its outer CGN layers remain intact at strains below 7.5%, while crack initiation and propagation begin at a strain of 8%. As strain increases, crack propagation accelerates, ultimately leading to failure. These observations confirm that the interpenetration of PE nanofibrous scaffolds not only improves the ductility of the internal interpenetrating network but also redistributes stress in the outer CGN layers to enhance that of the outer CGN layers. However, further improvements in elongation are limited by the intrinsic ductility ceiling of the CGN. Compared with natural materials and typical synthetic materials, including non-technical ceramics, polymers, foams, and elastomers, GLPs occupy a unique domain combining high strength and high modulus (Figure 2E and Figure S18).

The hardness (H) and elastic recovery (W_e) of GLP-3 were further evaluated by nanoindentation (Figure 2F). The load-displacement curve of GLP-3 exhibits minimal hysteresis, with calculated H and W_e values of 1.1 GPa and 78.3%, respectively (Figure 2G). The H of GLP-3 is comparable to that of ceramics and exceeds polymers and certain metals, indicating its glass-like hardness. Simultaneously, the W_e of GLP-3 surpasses that of most polymers, metals, and ceramics, approaching the values of typical rubbers, demonstrating its rubber-like resilience.

To explore the role of nano-hybrid interpenetrating network in inhibiting relaxation, probe relaxation tests were conducted. When the probe applies a force (~ 10 nN) on the sample surface for a brief duration (~ 0.1 s), the normalized stress curves (Figure 2H) show that PE and CPI films exhibit significant relaxation. In contrast, abundant rigid inorganic particles inhibit plastic relaxation from highly crosslinked GOS hybrid networks and PE nanofibrous scaffold [34, 35], resulting in minimal relaxation for CGN and GLP.

2.3 | Foldability of GLP

The foldability of GLP for practical applications was evaluated through dynamic and static folding tests. By combining its significantly enhanced ϵ_b and thin thickness, GLP shows

tremendous potential in both types of folding tests (Figure S19). To demonstrate their extreme foldability, the dynamic r_c was set to 0.5 mm (Figure 3A). Under this condition, commercially available foldable cover windows, such as CPI films, exhibit significant failure. The 30 μ m CPI film, owing to its poor creep resistance, develops visible creases in the folding area after just 100 folding cycles at $r_c = 0.5$ mm (Figures 3B,C), severely compromising the appearance and user experience of foldable displays. In contrast, GLP-3 maintains smooth surface and structural integrity even after 5 00 000 folding cycles (Figures 3D,E, Figure S20A, and Movie S1). Furthermore, the chemical structure, surface roughness, and modulus of the bent area of GLP-3 remain unchanged (Figure S20B–G, and Table S2). Meanwhile, when manufactured with a thickness of 30 μ m, GLP-3*1 and GLP-3*2 crack in the folding area (Figure S21), yet GLP-3*3 withstands 5 00 000 folding cycles at $r_c = 0.5$ mm without macroscopic creasing/cracking or microscopic morphology changes (Figure 3F). In static folding tests, GLP-3 withstands harsh temperatures (-20 and 80°C) with a r_c of 0.5 mm for 144 h without creasing or cracking (Figure 3G,H). The highly crosslinked GOS hybrid network and abundant rigid inorganic particles effectively inhibit irreversible strain (i.e., plastic relaxation and fatigue) during folding, preventing visible creases and ensuring the long-term reliability of foldable displays. Compared with state-of-the-art protective foldable materials, including flexible yet hard coatings and CPI [13, 16–19, 21, 38–43] GLP not only exhibit sufficient hardness but also demonstrate super-foldability, characterized by a low r_c and excellent durability over multiple folding cycles (Figure 3I,J, and Table S3). Additionally, GLP can also withstand twisting mode besides uniaxial folding, broadening its application in flexible electronics (Figure S22).

2.4 | Comprehensive Properties of GLP

Beyond foldability, comprehensive properties such as impact and scratch resistance, effective barrier properties against water vapor and gases, thermal stability, and comprehensive stability are crucial for practical applications. To assess impact resistance, ball drop tests were conducted (Figure 4A). Compared with UTG and CGN, GLP-3 exhibits the highest ball drop height, indicating superior impact resistance (Figure 4B–D and Figure S23). Specifically, the ball drop height of GLP-3 is 8.2 and 2.9 times higher than that of UTG and CGN, respectively. Visually, under drop hammer impact, UTG shatters violently after a single hammer strike (Figure 4E and Movie S2), whereas GLP-3 remains intact even after multiple impacts (Figure 4F and Movie S3). Meanwhile, GLP can be applied as a protective material on UTG to enhance its impact resistance, demonstrating its effectiveness in mitigating impact-induced damage in brittle substrates (Figure S24). This excellent impact resistance can be attributed to the highly crosslinked and nano-hybrid interpenetrating network in GLP (Figure 4G). Upon impact, the soft phase (organic phase) rapidly deforms to dissipate energy and transfers the concentrated stress to the PE phase, where the tightly entangled ultrahigh molecular weight PE fibrous scaffold effectively dissipates impact energy. When a crack is initiated in the CGN phase, the interpenetrating PE fibril networks could reduce (shield) the local stresses and strains actually experienced at the crack tip—stresses and strains that would otherwise extend

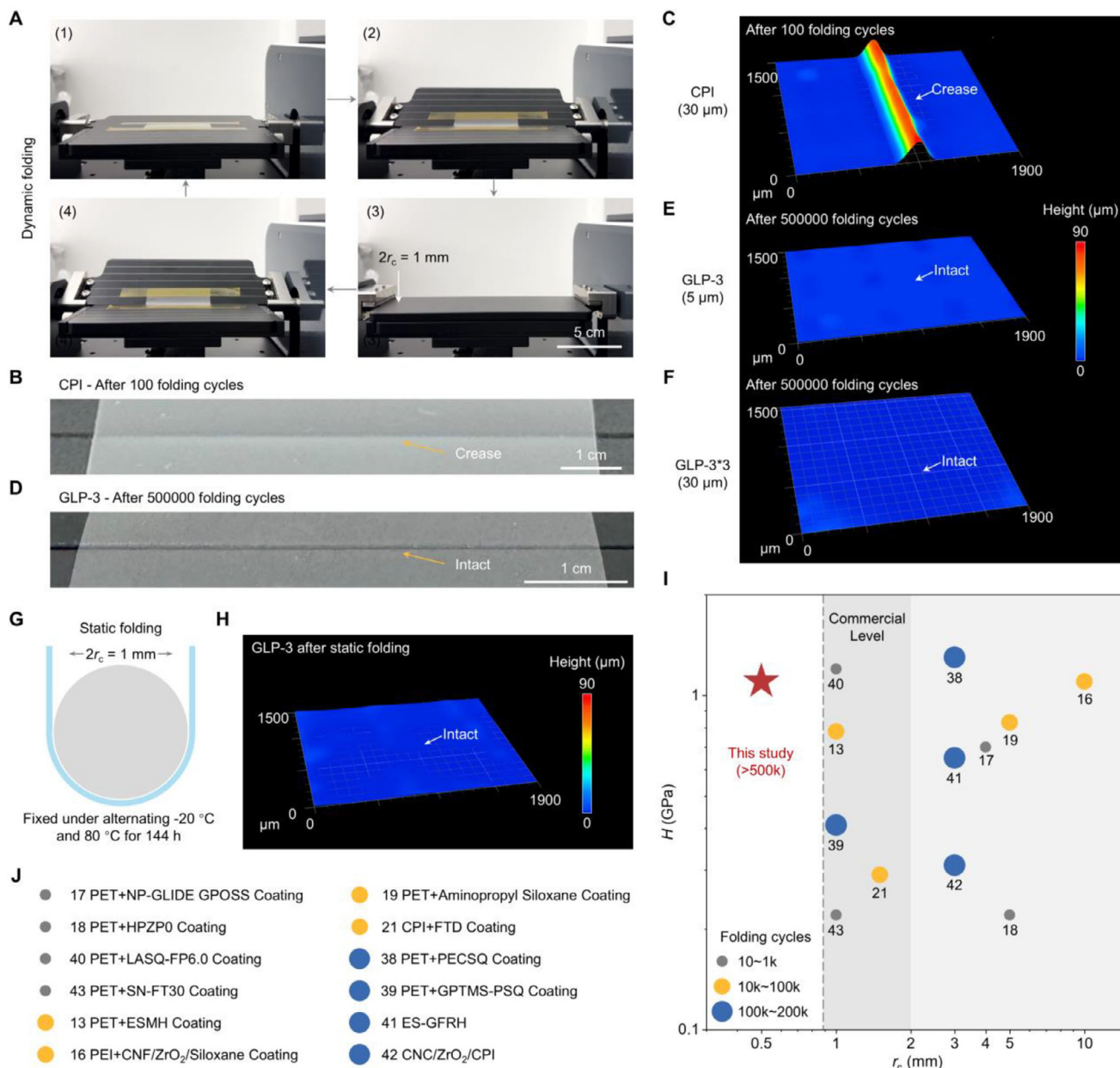


FIGURE 3 | Foldability of GLP. (A) photograph of dynamic folding at $r_c = 0.5 \text{ mm}$. (B, C) Photograph (B) and super deep field scene image (C) of 30 μm CPI films after 100 folding cycles at $r_c = 0.5 \text{ mm}$. D–E Photograph (D) and super deep field scene image (E) of 5 μm GLP-3 after 500 000 folding cycles at $r_c = 0.5 \text{ mm}$. It should be noted that all photographs were taken under oblique illumination to enhance the visibility of the highly transparent films. F Super deep field scene image of 30 μm GLP-3*3 after 500 000 folding cycles at $r_c = 0.5 \text{ mm}$. G Illustration of the static folding tests. H Super deep field scene image of 5 μm GLP-3 subjected to static folding at -20–80 °C for 144 h at $r_c = 0.5 \text{ mm}$. I–J Plot of H versus r_c for GLPs and the state-of-the-art foldable yet hard materials [13, 16–19, 21, 38–43]. The size of symbols represents different reported folding cycles.

the crack, achieving excellent impact resistance (Figure 4H and Figure S25).

Subsequently, the scratch resistance of GLP-3 was evaluated through steel wool wear tests (Figure 4I). SEM images reveal visible scratches on CPI films after only 100 wear cycles, whereas GLP-3 exhibits no discernible scratches even after 2500 wear cycles under a vertical pressure of approximately 30 kPa (Figure 4J,K). Post-wear surface properties, including water and oil contact angles and surface energy, remain consistent with the original values (Figure S26A). Moreover, GLP demonstrates

a friction coefficient comparable to UTG, lower roughness, and a smooth surface, along with glass-like surface hardness (8~9H pencil hardness) (Figure S26B–F and Table S4), collectively contributing to the excellent scratch resistance. Furthermore, GLP-3 also exhibits effective barrier properties against water vapor and gases (oxygen and carbon dioxide) and excellent thermal stability (Figure 4L, Figure S26G–I, and Table S5), making it an ideal material for electronic displays. To evaluate comprehensive stability, GLP-3 was treated with high-temperature, cyclic thermal shocks, UV exposure, 85°C/85% RH, and immersion in various solvents (organic solvents, acidic/alkaline solutions).

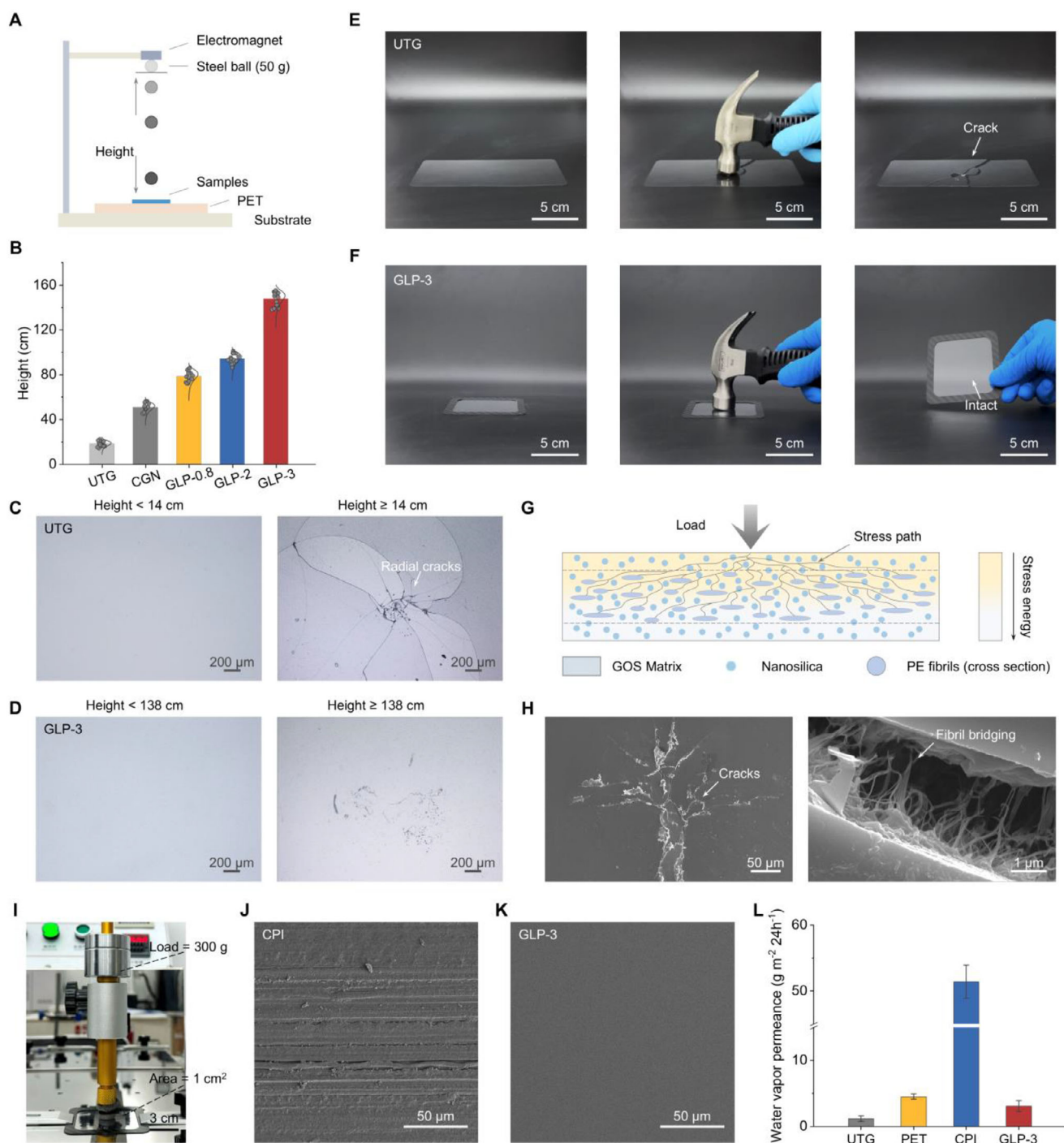


FIGURE 4 | Comprehensive properties of GLP. (A) Illustration of the ball drop tests. (B) Ball drop heights for UTG and GLPs with varying porous PE nanofibrous scaffold thicknesses. C–D Optical images of UTG (C) and GLP-3 (D) after ball drop tests with different drop heights. (E, F) Drop hammer tests for UTG (E) and GLP-3 (F). It should be noted that all photographs were taken under oblique illumination to enhance the visibility of the highly transparent films. (G) Illustration of the dissipation of stress energy for GLP. (H) SEM images of GLP-3 fracture surfaces. (I) Photograph of the steel wool wear test apparatus. (J, K) SEM images of CPI films (J) and GLP-3 (K) after 100 and 2500 abrasion cycles, respectively. L Water vapor permeance of UTG, PET, CPI, and GLP-3.

Because the highly crosslinked GOS hybrid network and rigid inorganic nanoparticles (CGN phase) provide effective oxygen/solvent barrier properties, excellent UV resistance, aging resistance and solvent resistance, GLP-3 retains its transmittance, surface morphology, roughness, tensile properties, and yellow index compared to original values, demonstrating robustness

under long-term harsh conditions (Figures S27 and S28, and Tables S6–S8). Additionally, to evaluate biological safety, we conducted in vitro assessments using extract leaching tests and direct-contact cytotoxicity assays, supplemented by live/dead staining (Figure S29). The extract assays reveal no detectable cytotoxic substances, with cells exhibiting normal proliferation

in GLP-3 extracts. Meanwhile, the direct-contact assays confirm excellent cytocompatibility, as cells proliferate normally on the GLP-3 surface. These results substantiate GLP's safety for flexible electronics.

The construction of the nano-hybrid interpenetrating network combines high transparency, excellent impact and scratch resistance, barrier and photothermal properties, and, more importantly, super-foldability for extreme dynamic and static folding conditions, leading to the generation of GLP for present foldable displays. Beyond foldable displays, the proposed GLP also holds promise for emerging applications such as wearable electronics, flexible sensors, and conformable optoelectronic systems, where the combination of mechanical robustness, optical clarity, and foldability is equally critical. The properties of such materials surpass those of current foldable protective materials (Table S3). Moreover, we anticipate its production cost will be lower than state-of-the-art UTG and CPI, owing to the relatively low cost of its main constituents: PE, GOS, and nanosilica (Supplementary Text, [Supporting Information](#)).

3 | Conclusion

In summary, GLP thin films are fabricated by infiltrating a GOS@nanosilica dispersion, with a precisely tuned refractive index, into a highly stretched, robust, porous ultrahigh molecular weight PE nanofibrous scaffold, followed by pressing and UV irradiation. The highly crosslinked GOS@nanosilica hybrid phase imparts high hardness and modulus, while inhibiting the fatigue relaxation of the plastic phase, whereas the porous PE nanofibrous scaffold enhances elongation and impact resistance. Owing to the unique nano-hybrid interpenetrating network structure and hybrid composition, our materials surpass the boundary between soft and hard materials, exhibiting glass-like transparency and hardness, plastic-like elongation, rubber-like elastic recovery and, more importantly, excellent impact resistance and super-foldability ($>5\,000\,000$ folding cycles at $r_c = 0.5$ mm) without macroscopic creasing/cracking or microscopic structure/morphology changes compared to existing materials, even under the harsh environmental conditions. With facile manufacturability, our study offers a straightforward methodology for integrating mutually exclusive functionalities into a single material, presenting substantial opportunities for future foldable and resizable displays.

4 | Experimental Section

4.1 | Materials

3-glycidyloxypropyl trimethoxysilane (Adamas, 98%+), tetraethyl orthosilicate (Adamas, 99%+), ammonia (Chengdu Kelong Chemical Co. Ltd., 25%–28%), hydrochloric acid (Chengdu Kelong Chemical Co. Ltd., 36%–38%), ethanol (Chengdu Kelong Chemical Co. Ltd., A.R.), methanol (Chengdu Kelong Chemical Co. Ltd., A.R.), toluene (Chengdu Kelong Chemical Co., Ltd., 99.9%+), hexane (Chengdu Kelong Chemical Co. Ltd., A.R.), triarylsulfonium hexafluoroantimonate (TSHFA) salts (Adamas, 50 wt.% in propylene carbonate), ultrahigh molecular weight PE resin with viscosity-average molecular

weight grade of 11 million g mol^{-1} (SG4850, Guangdong Suguang Plastic Co. Ltd), white Protobet 1S petrolatum (molecular weight ranging from 350 to 650 g mol^{-1} , Sonneborn), UTG (30 μm , TOMI, China), and CPI films (30 μm , Kolon, Korea). All chemicals were used as received without further purification.

4.2 | Preparation of Porous PE Nanofibrous Scaffolds

A uniform suspension was first prepared by heating and stirring a mixture of PE resin (5 wt.%) and petrolatum (95 wt.%) at 100°C. The resulting suspension was subsequently extruded into precursor gel sheets at 190°C using a twin-screw extruder (HAAKE Rheomex PTW 16 OS). These sheets were then biaxially stretched with a KARO IV system (Brückner, Germany). The initial stretching was conducted at 120°C with a stretching rate of 2% s^{-1} , yielding nanofibrous scaffolds with stretch ratios of 7.5×7.5 (56, PE-3) and 9×9 (81, PE-2). A secondary stretching was performed at 130°C with a stretching rate of 0.7% s^{-1} , generating a PE nanofibrous scaffold with final stretch ratios of $9 \times 9 \times 1.5 \times 1.5$ (182, PE-0.8). Finally, Soxhlet extraction with hexane was employed to remove the white petrolatum, producing porous PE nanofibrous scaffold with thicknesses of approximately 0.8, 2, and 3 μm .

4.3 | Synthesis of GOS

The title compound was synthesized following the procedure described in Figure S5 through the hydrolysis of 3-glycidyloxypropyl trimethoxysilane, as outlined in prior work with slight modifications [44, 45]. Specifically, 30 mL of 3-glycidyloxypropyl trimethoxysilane, 60 mL of hydrochloric acid, and 500 mL of ethanol were combined in a round-bottom flask, and the reaction mixture was heated to 60°C under vigorous stirring. After 24 h, the reaction was quenched, and GOS was obtained as a clear, viscous liquid following solvent removal.

4.4 | Preparation of GOS@nanosilica Dispersion

Nanosilica particles (diameter ~ 20 nm) were initially prepared using the Stöber method as described [46]. After purification by repeated centrifugation and washing with ethanol, the particles were stored as a stable dispersion in ethanol. Prior to use, the dispersion was concentrated, and the solvent was exchanged to toluene via multiple redispersion cycles. Subsequently, the nanosilica (10.0 g) dispersion in toluene (60 mL) was mixed with 3-glycidyloxypropyl trimethoxysilane (2.0 g). The mixture was stirred with a magnetic stirrer for 24 h to facilitate the reaction between the silane and hydroxyl groups on the silica surface. Afterward, the surface-treated nanosilica particles were washed thoroughly multiple times with toluene and methanol to ensure complete removal of unreacted silane. Finally, 14.0 g of GOS and 6.0 g of the surface-modified nanosilica particles were mixed in 20 mL of toluene under stirring. After solvent removal, the GOS@nanosilica dispersion was obtained as a highly viscous and clear liquid.

4.5 | Preparation of GLP

The porous PE nanofibrous scaffold was initially placed on a trichlorooctadecylsilane-treated glass, followed by infiltration with a dispersion of synthesized GOS@nanosilica mixed with 2.5 wt.% TSHFA salts. The samples were then covered with another trichlorooctadecylsilane-treated glass and subjected to pressing at a pressure of 0.5 MPa and a temperature of 80°C for 30 min to ensure uniform distribution and complete infiltration of the dispersion into the porous PE nanofibrous scaffold. After that, the samples were exposed to UV light in the UV-A region for 10 min to obtain GLP. GLPs with different porous PE nanofibrous scaffold thicknesses were fabricated using a similar process. GLP-3*1, GLP-3*2, and GLP-3*3 (30 µm) were fabricated based on GLP-3 using a similar process, by increasing overall film thickness and the number of porous PE nanofibrous scaffold layers.

4.6 | Scanning Electron Microscopy (SEM)

SEM images were captured on a Nova NanoSEM450 SEM (FEI, America) at an operating voltage of 10 kV. Prior to SEM analysis, all specimens were coated with a thin gold layer to mitigate the charging effect prior SEM analysis. The diameters of PE fibrils were quantified using ImageJ software. The corresponding EDS 2D mapping of C, O, and Si elements was recorded using the same equipment.

4.7 | Density Measurements

The densities of the samples were measured using a density analyzer (Matsuhaku, Japan) based on Archimedes principle. The mass of each specimen was recorded both in air and in deionized water, and the density was calculated accordingly. All measurements were conducted at room temperature, and the reported values represent the average of three measurements. The densities of GLP and CGN are measured to be 1.456 and 1.472 g cm⁻³, respectively.

4.8 | Atomic Force Microscope (AFM)

AFM height images were acquired under ambient conditions (room temperature, ~25°C; relative humidity, ~40%) using a Dimension Icon instrument (Bruker, America) in tapping mode. Modulus mappings were obtained using the same instrument in Peak-force QNM mode. The probe utilized was a RTESPA-525-30, with a corrected spring constant of 311.8 N m⁻¹ and a corrected tip radius of curvature of 31.0 nm. Probe relaxation tests were conducted using the same instrument. After initially contacting the sample surface, an input force (~10 nN) was applied for a brief duration (~0.1s) by the probe, and the indentation depth was maintained. The dependence of force on time was then recorded. Calibration of probe relaxation was done using a standard sapphire before each test. Image analysis was performed using NanoScope Analysis software, version 2.0.

4.9 | NMR

Liquid-state ¹H, ¹³C, and ²⁹Si NMR spectra of all samples were recorded in CDCl₃ using a 600 MHz Avance II NMR spectrometer (Bruker, America) at ambient temperature. The ²⁹Si NMR spectrum of GOS exhibits only two species, T² (−56 ~ −61 ppm) and T³ (−65 ~ −72 ppm). Equation 1 was used to calculate the degree of condensation (DOC) of the GOS for quantitative analysis. The degree of condensation of the GOS was 97.9%, suggesting that the siloxane framework of the GOS is highly condensed through the sol-gel reaction.

$$DOC (\%) = \frac{T^1 + 2T^2 + 3T^3}{3(T^0 + T^1 + T^2 + T^3)} * 100 \quad (1)$$

Solid-state ²⁹Si NMR spectrum was recorded on a 500 MHz Avance III NMR spectrometer (Bruker, Sweden).

4.10 | Size Exclusion Chromatography (SEC)

The molecular weight of GOS was determined via SEC using tetrahydrofuran as the eluent. The analysis was performed on a Waters 1515 instrument (Waters, America) equipped with a gel column, a differential refractive index detector, and a light scattering detector. The system was operated at a sample concentration of 2 mg mL⁻¹ and a solvent flow rate of 0.5 mL min⁻¹. Standard polystyrene samples were employed for calibration of molecular weight and polydispersity calculations.

4.11 | Matrix-assisted laser desorption ionization time-of-flight mass spectrum (MALDI-TOF MS)

MALDI-TOF MS was acquired using an AXIMA Performance MALDI TOF/TOF mass spectrometer (Shimadzu, Japan) in positive reflection mode. *Trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB, TCI, >99%) served as the matrix and was dissolved in chloroform at a concentration of 20 mg mL⁻¹. Sodium trifluoromethanesulfonate (NaTFA, Adamas, 98%) was used as the cationizing agent and dissolved in a 1/3 (v/v) mixture of methanol and chloroform at a concentration of 10 mg mL⁻¹. Samples were prepared by dissolving them in chloroform at a concentration of 5 mg mL⁻¹. For measurement, 0.5 µL of the matrix and cationizing agent solution mixture was deposited onto the wells of a 384-well ground-steel plate and allowed to dry. Then, 0.5 µL of each sample was deposited onto the dried matrix spot, followed by another 0.5 µL of the matrix and cationizing agent solution mixture.

4.12 | Transmission Electron Microscopy (TEM)

Bright-field TEM images were obtained using a Talos F200i microscope (Thermo Fisher Scientific, America) at an operating voltage of 200 kV. To prevent aggregation of nanosilica, a diluted suspension of nanosilica was deposited on an ultrathin carbon film. The average diameter of nanosilica particles was measured by counting over 200 particles using ImageJ software. For internal

structure characterization, GLP-3 was microtomed to a thickness of 50–60 nm for TEM observation. HAADF-STEM image and corresponding EDS 2D mapping of C, O, and Si elements were recorded using the same equipment.

4.13 | Rheological Test

Rheological tests were carried out at room temperature ($\sim 25^{\circ}\text{C}$) using a rheometer (MCR-302, Anton-Paar, Austria) equipped with a 25 mm diameter parallel steel plate over a shear rate range of $0.1\text{--}100\text{ s}^{-1}$.

4.14 | Contact Angle

Contact angles were measured using a DSA25 instrument (KRÜSS, Germany). The surface energy of the samples was calculated based on the contact angle measurements of water and diiodomethane, employing the Owens-Wendt-Rabel-Kaelble (OWRK) method [47]. For each measurement, 2 μL droplets were applied, and each test was repeated at least three times across different areas of the sample. It should be noted that nonporous PE films were fabricated by hot-pressing ultrahigh molecular weight PE resin.

4.15 | Refractive Index

Refractive index spectra were measured using an IR-VASE Mark II M-2000UI instrument (J. A. Woollam, America) over a wavelength range of 200–1600 nm. Given the difficulty in measuring the refractive index of the PE nanofibrous scaffold, a nonporous PE film was used. Meanwhile, the in-plane refractive indices (at 0° , 45° , and 90°) and out-of-plane refractive index of GLP-3 were measured using an RC2 instrument (J.A. Woollam, America) over a wavelength range of 400 to 1000 nm. Notably, due to differences between instruments, the refractive indices show slightly difference from CGN and GLP-3.

4.16 | Ultraviolet-Visible Spectroscopy (UV-Vis)

UV-vis spectra were carried out under ambient conditions (room temperature, $\sim 25^{\circ}\text{C}$; relative humidity, $\sim 40\%$) using a Lambda 650 spectrophotometer (Perkin Elmer, America) over a wavelength range of 200–800 nm, with air serving as the baseline. It should be noted that the optical transmittance is specular (parallel) transmittance. The haze values were measured on a lambda 750 UV-Vis-NIR spectrophotometer (Perkin Elmer, America) with a wavelength range of 400–800 nm.

4.17 | Fourier Transform Infrared Spectroscopy (FT-IR)

FTIR spectra were recorded on a 6700 spectrometer (Nicolet, America) in transmission and attenuated total reflection model over the wavenumber range of $650\text{--}4000\text{ cm}^{-1}$ with a resolution of 4 cm^{-1} . The FT-IR spectra of GLP at varying UV curing times (0–10 min) and after 24 h of ambient storage post-curing were

monitored, and the conversion of epoxy groups was analyzed to confirm the completion of the crosslinking reaction.

4.18 | Wide-Angle X-Ray Diffraction (WAXD)

WAXD patterns were obtained using an X'pert pro MPD X-ray diffractometer (PANalytical B.V., Netherlands), employing an incident X-ray wavelength of $1.54\text{ \AA}$ (Cu $K\alpha$ radiation). The instrument was operated at a voltage of 40 kV and a current of 40 mA. The scanning range for 2θ was set from 5° to 90° , and the scan rate was $10^{\circ}\text{min}^{-1}$.

4.19 | Stylus-Type Profilometry

The thickness of the samples was measured using a Dektak XT stylus profilometer (Bruker, America). Each sample was partially mounted on a silicon wafer, creating a step with the wafer serving as the baseline. Scans were across the samples and the silicon wafer, with a scan length of $500\text{ }\mu\text{m}$ and a scanning rate of $100\text{ }\mu\text{m s}^{-1}$. Line profiles of the samples' surface were recorded using the same instrument, with a scan length of $2000\text{ }\mu\text{m}$ and a scanning rate of $100\text{ }\mu\text{m s}^{-1}$. Notably, to eliminate sample inclination, the line profiles were flattened.

4.20 | Tensile Tests

Tensile tests were conducted under ambient conditions (room temperature, $\sim 25^{\circ}\text{C}$; relative humidity, $\sim 40\%$) using an EM6.501-W universal testing machine (Shenzhen TestMate Instruments & Equipment, China) at a strain rate of $50\%\text{ min}^{-1}$. Each specimen, cut to dimensions of $10\text{ mm} \times 15\text{ mm}$, was firmly clamped between grips on the universal tensile tester. Notably, GLP-3*3 after undergoing fixed strain treatment was prepared by pausing the tensile process and holding for 3 min.

4.21 | Nanoindentation

Nanoindentation analysis was carried out under ambient conditions (room temperature, $\sim 25^{\circ}\text{C}$; relative humidity, $\sim 40\%$) using a TI 980 nanoindenter (Bruker, USA) equipped with a diamond Berkovich tip. Calibration was performed utilizing standard samples before conducting the tests. The test procedure involved 5 s of loading to 10 mN and 5 s of unloading. Hardness and elastic recovery of samples were calculated from the recorded loading force and indentation depth using the Oliver-Pharr method [48, 49]. Notably, GLP-3 samples used for nanoindentation tests were cemented onto quartz glass before UV curing. Additionally, the thickness between the PE nanofibrous scaffold and quartz glass was increased to ensure that the sample thickness on the quartz glass surface exceeded $100\text{ }\mu\text{m}$ to eliminate the effect of the substrate.

4.22 | Dynamic Folding Tests

Dynamic folding tests were performed under ambient conditions (room temperature, $\sim 25^{\circ}\text{C}$; relative humidity, $\sim 40\%$) using a

KZ-FD2 folding cycle tester (Kezhizhu Electronic Technology Chongqing Co., Ltd, China). The samples were subjected to repeated folding into a U-shape and recovered to a flat state, with a r_c of 0.5 mm. Observations reveal that the UTG (30 μm) exhibits cracking after one folding cycle, while CPI films (30 μm) develop creases after 100 cycles of folding. In contrast, no cracks or creases were detected in the folding areas of GLP, even after 5 00 000 folding cycles. The surface morphology of the folding areas was examined using a Smartzoom 5 super deep scene 3D microscope (ZEISS, Germany).

4.23 | Static Folding Tests

For static folding tests, the samples were folded into a U-shape and fixed onto a cylindrical steel rod with a diameter of 1 mm. They were then exposed to alternating temperature variations between -20 and 80°C for 144 h, with temperature switching every 12 h.

4.24 | Impact Resistance

Impact resistance was evaluated using a ball drop test. A 50 g steel ball with a diameter of 23 mm was dropped from a predetermined height onto the test specimen, which was positioned on a PET/steel substrate. After each drop, the surface morphology of the specimen was examined, and the height at which damage first occurred was recorded. The impact resistance was quantified by determining the maximum height from which the ball could be dropped without causing damage to at least 15 different regions of the samples. Notably, the GLP-coated UTGs were fabricated using a process similar to that of GLP, with the exception that the bottom trichlorooctadecylsilane-treated glass was replaced by UTG. The damaged area was captured using an optical microscope. Micro-CT tests were performed using a ZEISS Xradia 610 Versa X-ray microscope, data were collected at 80 kV using a $20\times$ objective lens, and the 3D microstructures were reconstructed using the Dragonfly software package.

4.25 | Scratch Resistance

Scratch resistance was tested under ambient conditions (room temperature, $\sim 25^\circ\text{C}$; relative humidity, $\sim 40\%$) using a ZOT-6014 339 abrasion tester (Guangdong Zhongtian Instrument Co., Ltd, China). For each test, fresh steel-wool (grade #0000, fiber diameter of 25 μm) was tightly wrapped and securely fixed to the tester. A vertical pressure of approximately 30 kPa (with a contact area of $\sim 1\text{ cm}^2$ and a load of 300 g) was applied, with an abrasion frequency set to 1 Hz.

4.26 | Friction Coefficient

The friction coefficients were obtained from nano-scratch tests using a TI 980 nanoindenter (Bruker, USA) with an applied load of 500 μN and a scratching distance of 15 μm .

4.27 | Water Vapor Permeance

The water vapor permeances of the samples were measured using a PERME W3/031 tester (Labthink, China), based on the moisture permeation cup weighing method, at a testing temperature of 37°C and a humidity of 80%.

4.28 | Gaseous-Diffusion Coefficient

The gaseous-diffusion coefficients of the samples were determined using a VAC-V2 tester (Labthink, China), based on the differential pressure testing principle.

4.29 | Pencil Hardness

Pencil hardness testing was conducted under ambient conditions (room temperature, $\sim 25^\circ\text{C}$; relative humidity, $\sim 40\%$) using an Elecometer 501 pencil hardness tester, following the ASTM D3363 protocol. Standard pencils (Mitsubishi, Japan), with hardness grades ranging from 6B (softest) to 9H (hardest), were applied at a 45° angle to the sample surface under a load weight of 0.75 kg and then moved across the surface. Pencil hardness was determined by the hardest pencil that did not cause any damage to the surface in at least five different areas.

4.30 | Thermogravimetric Analysis (TGA)

TGA analysis was performed using a TG 209 (Netzsch, Germany) under both nitrogen and air atmospheres. The temperature range was from room temperature to 800°C with a heating rate of $10^\circ\text{C min}^{-1}$.

4.31 | Dynamic Mechanical Analysis (DMA)

DMA was performed using a TA Q800 (America) with a frequency of 1 Hz. The temperature range was from -50 to 250°C with a heating rate of 5°C min^{-1} .

4.32 | Comprehensive Stability

The comprehensive stability of the samples was evaluated by treating them with high temperature (100°C , 25 days), cyclic thermal shocks (100 cycles between -20 and 100°C), UV exposure (270 W m^{-2} , 25 days), $85^\circ\text{C}/85\%$ RH (15 days), and immersion in various solvents (ethyl alcohol, ethyl acetate, toluene, saturated brine, 1 M HCl, and 1 M NaOH, 7 days).

4.33 | Cell Culture

NIH 3T3 cells were obtained from Sichuan University and cultured in Dulbecco's Modified Eagle Medium (DMEM, Gibco-BRL, Grand Island, NY, USA) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin

(10 000 U/mL penicillin and 10,000 µg/mL streptomycin). Cells were maintained at 37°C in a humidified incubator containing 5% CO₂.

4.34 | Cell Viability

The cell viability was evaluated using the standard Methyl Thiazolyl Tetrazolium (MTT) assay. (a) Extract leaching method: Film samples were sterilized by ethanol immersion and then incubated in complete cell culture medium at 37°C for 24 h. After incubation, the supernatant (2 mL) was collected as the extract solution. NIH 3T3 cells were seeded in 96-well plates at a density of 2,000 cells per well and incubated for 24 h. The culture medium was then replaced with the extract solution, and the cells were further incubated for 24 and 72 h. Cell viability was then assessed using the MTT assay. (b) Direct-contact method: Film samples were first sterilized by ethanol immersion and placed into the wells of a 12-well plate. Subsequently, 100 µL of the NIH 3T3 cell suspension was carefully dropped directly onto the surface of each film. The plates were incubated for 30 min. After that, 2 mL of complete culture medium was gently added to each well. The cells were further incubated for 24 and 72 h. After incubation, cell viability was assessed using the MTT assay.

4.35 | Calcein-AM and Propidium Iodide Staining

Calcein-AM and propidium iodide staining assays were performed to further evaluate cytocompatibility. Briefly, NIH 3T3 cells were seeded in 6-well plates at a density of 2 00 000 cells per well and co-incubated with the film samples for 24 and 72 h, either by extract or direct contact method. After incubation, cells were stained with Calcein-AM and propidium iodide (PI) for 15 min at 37°C in the dark. The stained cells were then imaged using a fluorescence microscope (Carl Zeiss, Germany).

4.36 | Statistical Analysis

In this study, statistical data are presented as means ± standard deviations or, in some cases, the means of all results. Means and standard deviations were calculated using the Mean and STDEV.P functions in Microsoft Excel.

Author Contributions

Conceptualization: X.L., D.H., Q.F.; Methodology: X.L., D.H., Q.F.; Investigation: X.L., Y.W., R.L., J.Y., D.-L.Z., C.-Y.L., Z.W., J.W., D.H., Q.F.; Visualization: X.L., D.H., Q.F.; Funding acquisition: D.H., Q.F.; Project administration: D.H., Q.F.; Supervision: Q.Z., F.C., D.H., Q.F.; Writing – original draft: X.L., D.H., Q.F.; Writing – review & editing: X.L., D.H., Q.F.

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Conflicts of Interest

D.H., X.L., Q.F., Y.W., C.-Y.L., and Q.Z. applied for three Chinese patents (application nos. 202510154760.6, 202510852946.9, 202510752776.7). The authors declare no other competing interests.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adma73424-sup-0001-SuppMat.pdf.

Supporting File 2: adma73424-sup-0002-MovieS1.mp4.

Supporting File 3: adma73424-sup-0003-MovieS2.mp4.

Supporting File 4: adma73424-sup-0004-MovieS3.mp4.